



ONE SHOT

REVISION SERIES

CLASS 10 **GEOGRAPHY**

MANUFACTURING INDUSTRIES



Introduction

Industries

⇒ [Secondary sector] ⇒ Raw material to finished goods.

Importance of manufacturing : **[Backbone of development]**

- It reduce the dependence of people on agriculture income. [By providing employment]
- Industrial development, a precondition for eradication of unemployment and poverty. [Public sector investment]
- Export brings much needed foreign exchange.
- Manufacturing provides higher value of good to a country. [Prosperity]

Agriculture and Industry are not exclusive of each other.



Moves hand in hand



Challenges given by Globalisation. 🤖

Classification of Industries

On the basis of source of raw materials

- Agro based
- Mineral based

According to their main role -

- Basic or key industries
- Consumer Industries

On the basis of capital investment

- Small scale industries
- Large scale industries

On the basis of ownership

- Public sectors
- Private sector
- Joint sector
- Cooperative sector

Based on the bulk and weight of raw material and finished goods

- Heavy industries
- Light industries

Agro Based Industry: Textile Industry

Industries based on agricultural raw material

Cotton, jute, silk, woollen textiles,
sugar and edible oil, etc

Textile Industry:

- Contribution in industrial production
Employment generation
- Foreign exchange earnings
- Only industry in the country, which is self-reliant and complete in the value chain

Cotton Textile



Jute Textile



Cotton Textile

- First successful cotton mill was established in Mumbai in 1854.
- Hand spinning and hand looms weaving.
After the 18th century, **power -looms** came into use 🤔
- Localisation of textile industries in Gujarat and Maharashtra. 🤔
- Availability of raw cotton, market, transport [port], labour moist climate.

**Textile Industries**

- Association with other Industries [Chemical and dye, mill store and packaging]
- Link with farmers and workers.



Spinning concentrated in Maharashtra, Gujarat and Tamil Nadu while weaving is decentralized to incorporate traditional skills.

Challenges to cotton textile industries

- Erratic power supply.
- Upgradation of machinery.
- Low output of labour.
- Stiff competition with the synthetic fibre industries.

Jute Textiles

- Largest producer of raw jute and second largest exporter after Bangladesh.



Problems after partition 1947

Most of the jute mills are located in West Bengal, near Hugli River.

1. Proximity to jute producing area.
2. Inexpensive water transport.
3. Good network of railways.
4. Abundance of water for processing raw jute.
5. Cheap labour from West Bengal and adjoining areas.
6. Kolkata urban centre: Urban facilities.

First jute mill near Kolkata in 1859, at Rishra.



Sugar Industry

India



Second largest producer of sugar
[Largest producer of Gur and Khandsari].

Bulky raw material



In haulage its sucrose content reduces.



∴ Industries should be situated near sugarcane producing area.

[60% mill are situated in U.P. and Bihar]

- Seasonal in nature, ideally suited to cooperative industry? →
- Increase in number of mills in Southern and Western states because of favorable conditions. 🤔

Why



- The cane produced here has a higher sucrose content.
- The cooler climate also ensures a longer crushing season.
- Moreover, the cooperatives are more successful in these states.



Mineral Based Industry: Iron and Steel Industry

- It is basic industry. ➡ **Why**
- Dependence of other industries over it.
- Production and consumption of steel is often regarded as the index of a country's development
- Iron ore, coking coal and lime stone are required in the ratio of approximately 4 : 2 : 1.
- Chota Nagpur plateau has the maximum concentration of iron and steel industries. ➡ **Why**
 - 1. Low cost iron ore 2. High grade raw material 3. Cheap labour 4. Market

Aluminium Smelting

➡ Second most important metallurgical industry.



- Light, resistant to corrosion, good conductor of heat, malleable and becomes strong when mixed with metal.
- It is used to manufacture aircraft, utensils and wires.
- It has gained popularity as a substitute of steel, copper, zinc and lead in a number of industries
- Aluminium smelting plants in the country are located in Odisha, West Bengal, Kerala, Uttar Pradesh, Chhattisgarh, Maharashtra and Tamil Nadu.

4 to 6 tonnes of Bauxite**2 tonnes of Alumina****1 tonnes of Aluminium****Challenges:** Regular supply of electricity and assured source of raw material at minimum cost.

Chemical Industry

- It comprises of both large and small scale industries.
- Rapid growth in organic and inorganic chemical industries.
- It's own largest consumer.
 - Undergoes processing to further produced other chemicals.



Fertilizer Industry

- Most of the industries are centred around the production of nitrogenous fertilizers (urea) phosphatic fertilizers and ammonium phosphate (D.A.P.).
- Potash is imported in our country.
- Industry ↑ → After Green Revolution.
- Prominent in the areas of Gujarat, T.N., U.P. and Kerala.



Cement Industry

Essential for construction activity, building houses, factories, bridges, etc.

Requirement

- This industry required bulky and heavy raw material like limestone, silica, etc.
- Coal and electric power along with transportation. → [Railways]
- Situated in Gujarat and coastal area because of accessibility to Gulf countries.



Automobile Industry

- The industry provide base for transport [Goods and Passengers].
- E.g. Trucks, Buses, Cars, Motorcycles, etc.
- Liberalisation ➡ Competition ➡ Growth of Industry 🧐
- Foreign direct investment brought in new technology and aligned the industry with global developments.
- The industry is located around Delhi, Gurgaon, Chennai, Pune, Mumbai, Kolkata, Lucknow, Indore, Hyderabad, Jamshedpur and Bangalore.



Information Technology and Electronics Industry

- Wide range of products → Calculator to computer.
- Bangalore ➡ Electronic capital of India ➡ Why ➡ Concentration of industries
- Other centres are Delhi, Chennai, Pune, Mumbai, Kolkata, Lucknow, and Coimbatore.
- A major impact of this industry has been on employment generation.
- The continuing growth in the hardware and software is the key to the success of IT industry in India.

Industrial Pollution and Environmental Degradation

Industry contribute a lot in development but in turn it has caused pollution and environmental degradation.

Air Pollution

- Caused by the presence of high proportion of **undesirable gases** such as sulphur dioxide and carbon monoxide.
- **Airborne particulate materials** contain both solid and liquid particles like dust, sprays mist and smoke.
- Chemical and paper **factories burning** fossil fuels, bricks kilns, refinery and smelting, etc. leading to smoke.
- Leads to pollution and hazards to human health, plants, animals and atmosphere.

Water Pollution

- Caused by organic and inorganic **waste** and effluents.
- **Main culprits** - Paper, pulp, chemical, textile and dyeing, petroleum refineries, tanneries and electroplating industries.
- These **industries dump substance** like dyes, detergent, acids, salts and heavy metals like lead and mercury, pesticides and fertilizers, plastic and rubber.
- Fly-ash, phospho-gypsum, and iron and steel slags are the major **solid waste**.

Thermal Pollution

- When hot water from factories and thermal plant is drained into rivers and ponds it causes water pollution.

Impact :

- Waste dumped is highly toxic [Nuclear power plant] lead to concern,
- Birth defect and miscarriage.
- It harm the aquatic life.

Relation between soil and water pollution :

Dumping waste on soil. → Rain water percolates.

Groundwater gets contaminated.

Noise Pollution

- Caused due to industrial and constructional activities.
- Machinery, factory equipment, generators, saw, pneumatic and electric drills.
- Hospitals and Schools.

Impact :

- Results in irritation and anger, it can also cause hearing impairment, increased heart rate and blood pressure.
- Unwanted sound is source of stress and reason for lack of concentration.

Control of Environmental Degradation

Some suggestions to control Environmental Degradation

- Minimising use of water for processing by reusing and recycling it.
Rainwater harvesting to meet our requirement.
- Treating hot water and effluents before releasing them into water bodies.
Industrial effluents can be treated in three ways -
 - Primary treatment by mechanical means.
 - Secondary treatment by biological process.
 - Tertiary treatment biological, chemical and physical processes.
- Using of groundwater reserve by industries should be regulated legally.
- Using oil or gas instead of coal in factories to control smoke.
- Generators should be fitted with silencers.
- Noise absorbing material may be used apart from personal use of ear plugs and ear phone.

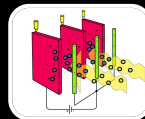
Control of Environmental Degradation

Particulate matter in air can be reduced by following methods:

- Fitting smoke stacks in factories with

- Electrostatic precipitators
- Fabric filters
- Scrubbers
- Inertial separators

Sustainable development
requires integration of economic
development with
environmental concern.



NTPC Shows the Way

- National Thermal Power Corporation has **ISO certification** for Environment Management System [EMS].
- NTPC has taken proactive approach for preserving the natural environment and resources.
 - Adopting latest techniques and upgrading equipment.
 - Minimising waste generating by maximising ash utilisation.
 - Providing green belts for nurturing ecological balance. → **[Afforestation]**

Step taken →

- Ash pond management, ash water recycling system and liquid waste management to reduce environmental pollution.
- Ecological monitoring, review and online data management for all its power stations.



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